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Paleomagnetism, stratigraphy, and petrology of the Upper Cretaceous Roskruge Volcanics at Bell Mountain, southeast Arizona

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Abstract. Reconnaissance geologic mapping and petrologic study of the Upper Cretaceous Roskruge Volcanics at Bell Mountain in the Roskruge Mountains, southeastern Arizona, have identified five compositionally distinct ash flow tuffs, including the basal red Viopuli Tuff, which comprise six or seven separate cooling units. Previous paleomagnetic investigators inferred eleven cooling units at Bell Mountain, but insufficient information was given to determine which tuffs had been sampled. This study was undertaken therefore to establish characteristic remanence directions for each tuff sheet. The available paleomagnetic data corroborate the identification of five map units in the Bell Mountain area, but the number of independent observations is too few for the calculation of a reliable paleopole; paleosecular variation of the geomagnetic field would probably not be averaged out. No correlatable tuffs are present at Bell Mountain, so relative vertical axis rotations cannot be determined, although they are probably not large. The absence of regionally correlatable tuffs suggests the presence of additional, potentially mineralized, volcanic centers perhaps farther north in the Roskruge Mountains or buried nearby beneath Cenozoic basin fill.

Introduction

Upper Cretaceous volcanic rocks of the Roskruge Mountains in southeastern Arizona were initially sampled for paleomagnetic study by *Vugteveen et al.* [1981] to determine a reference paleomagnetic pole (~72 Ma) for the North American plate. This paleopole is of particular interest because few poles are available for North America during the Laramide orogeny and corresponding interval of rapid apparent polar wander. The Roskruge Volcanics include a thick sequence of well-stratified outflow ash flow tuffs, containing many separate cooling units [*Bikerman*, 1967] that apparently constitute the best preserved section of Cretaceous tuff sheets in Arizona. This section was collected by *Vugteveen et al.* [1981] at Bell Mountain (Figure 1) on the Tohono O'odham Indian Reservation (access now restricted) at 11 sites, each interpreted to represent an individual cooling unit. Our geologic mapping and petrologic study (Table 1), however, has shown that only five major ash flow compositional units (6 or 7 probable cooling units) are exposed in the Bell Mountain area.

Site locations are not given by *Vugteveen et al.* [1981] in sufficient detail to relate their data to the newly identified tuffs. Also, our studies of the Cretaceous ash flow tuffs and associated caldera sources in the nearby Silver Bell, Tucson, and Sierrita Mountains [*Lipman and Sawyer*, 1985; *Sawyer*, 1989; *Hagstrum and Sawyer*, 1989; *Lipman and Fridrich*, 1990; *Hagstrum and Lipman*, 1991; *Lipman*, 1993] have raised the possibility that outflow tuff sheets from these calderas might be preserved in the Roskruge Mountains. The ash flow tuffs in the Bell Mountain area (Table 1) thus were resampled for paleomagnetic analysis (outside the reservation in low hills near the San Pedro Road) to establish their characteristic directions of remanent magnetization, thereby providing data relevant to possible correlations with regional ash flow tuffs and post-Late Cretaceous vertical axis rotations, and to assess the reliability of the previously published Late Cretaceous pole.

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Geologic Setting

The 800- to 1000-m-thick sequence of Upper Cretaceous ash flow tuffs in the Roskruge Mountains, the Roskruge Volcanics of *Bikerman* [1967], forms the main part of the mountain range. The lowermost tuff of the Roskruge Volcanics, the densely welded Viopuli Ignimbrite (herein called the Viopuli Tuff), has a K-Ar biotite age of ~76 Ma, and the overlying rhyolite and dacite tuffs (tuff units 1–4) have K-Ar biotite ages ranging from ~74 to ~70 Ma [*Bikerman*, 1967]. These upper tuffs are coeval with, and chemically similar to, the ~70 Ma Cocoraque Butte Granodiorite [*Bikerman*, 1967, 1968]. The Roskruge Volcanics unconformably overlie middle to Upper Cretaceous andesitic lava flows and interbedded clastic sediments of the Cocoraque and Roadside Formations [*Heindl*, 1965] and are succeeded and intruded by Tertiary rocks including the La Tortuga Butte Granodiorite (~35 Ma), basaltic andesite lava flows and associated sedi-

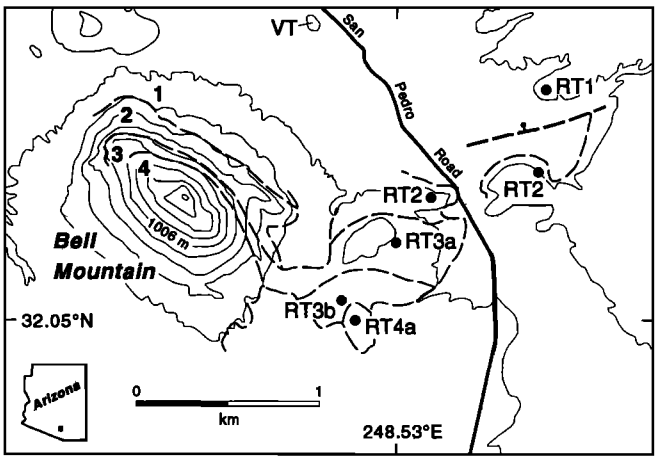


Figure 1. Reconnaissance geologic map of Bell Mountain and vicinity in the Roskruge Mountains showing locations of the paleomagnetic sampling sites for this restudy (see Table 1). Dashed lines show contacts, and the thicker dashed line east of the San Pedro Road indicates an inferred normal fault downthrown to the north. VT, Viopuli Tuff. The summit of Bell Mountain is at 1132 m, and the contour interval is 30.5 m.

ments (~24 Ma), the Recortado Tuff (~13 Ma), and basaltic dikes and flows (~10–15 Ma) [*Bikerman*, 1967, 1968].

Northeast of Bell Mountain (Figure 1), an isolated outcrop of densely welded red-brown crystal-poor dacitic tuff (Table 2), containing a few percent phenocrysts of biotite and plagioclase, appears to be the red Viopuli Tuff of *Bikerman* [1967]. This ash flow tuff is apparently overlain (unless complicated by faulting) by weakly indurated gray dacitic phenocryst-rich biotite tuff and tuffaceous sandstone (without quartz) which is poorly exposed on the pediment toward Bell Mountain. The gray tuffs are directly overlain by a dark gray (dacitic?) lava flow near the base of Bell Mountain. The lowermost ash flow sequence at the north base of Bell Mountain (tuff 1) consists of two compositionally similar cooling units of light gray phenocryst-rich quartz-biotite rhyolite (units 1a and 1b), each 10–15 m thick, separated by a few meters of bedded tuffaceous sandstone.

Overlying tuff 1 and approximately 10 m of dark gray volcaniclastic sandstone (indicating a major depositional break) is a massive ash flow sheet (tuff 2) of phenocryst-poor silicic rhyolite (Table 2). Tuff 2 is ~80 m thick on the north side of Bell Mountain and weathers orange brown; it is at most only partly welded and is characterized by abundant centimeter-size lithic fragments of flow-layered rhyolitic lavas. Tuff 2 contains about 5% small phenocrysts of mostly argillized feldspar and sparse quartz. Several ash flow subunits separated by discontinuous beds, interpreted as surge deposits within a single ash flow sheet, are present near the top of tuff 2, but no sedimentary interval was found between tuffs 2 and 3.

Tuff 3 is a distinctive cliff-forming brown welded tuff sheet characterized by large gray lenses of flattened pumice (≤30 cm long) and by the absence of quartz phenocrysts. Tuff 3 appears to be a single cooling unit on the north side of Bell Mountain where it is ~40 m thick, but it thins and becomes less welded to the east. Exposed in the low hills between Bell Mountain and the San Pedro Road (Figure 1) is a probable surge deposit (<1 m thick) along a partial welding break which defines subunits 3a and 3b; no compositional break is evident between these subunits although there is an apparent upward change in bulk composition from dacite to rhyolite (Table 2). Minor element compositions are relatively uniform for the samples analyzed, and the high Zr content is distinctive. Densely welded tuff of unit 3 contains about 10% large (2–3 mm) blocky phenocrysts of altered feldspar and biotite; pumice lenses have only a few percent phenocrysts. The upper part of tuff 3 becomes light gray and displays vapor-phase crystallization and silicification. No bedded tuff or sediment is exposed at the top of tuff 3, but the contact with tuff 4 is considered a complete cooling break between separate ash flow sheets because of an abrupt termination of vapor-phase crystallization.

The uppermost ash flows (tuff 4) on Bell Mountain are compositionally similar silicic rhyolite tuffs that show a complex emplacement and cooling history. These tuffs were interpreted in the field as a single compound cooling unit about 100 m thick, consisting of two gray-brown welded zones (4a and 4b) separated by an intervening nonwelded to partly welded yellow-tan interval. Both units of tuff 4 are compositionally distinctive silicic rhyolite, containing about 10% small broken crystals of quartz and argillized feldspar and characterized by highly fractionated minor element compositions (high Rb, Y, and Nb and low Ba and Sr; Table 2).

Table 1. Upper Cretaceous Ash Flow Tuffs of the Bell Mountain Area, Roskruge Mountains, Southeast Arizona

Unit	Composition	Magnetic Polarity	Sampling Sites: Vugteveen*	This Study
Tuff 4	crystal-poor silicic rhyolite			
Subunit 4b	upper cooling unit	N	RV11–12	...
Subunit 4a	lower cooling unit	R	RV6–10	RT4a
Tuff 3	pumice-rich rhyolite-dacite			
Subunit 3b	upper cooling unit	N	...	RT3b
Subunit 3a	lower cooling unit	N	...	RT3a
Tuff 2	lithic-rich silicic rhyolite	N	...	RT2
Tuff 1	crystal-rich low-silicic rhyolite	R	RV5†	RT1†
Subunit 1b	upper cooling unit			
Subunit 1a	lower cooling unit			
Viopuli Tuff	crystal-poor silicic dacite	N	RV4	...

*Inferred from *Vugteveen et al.* [1981].
†Uncertain which cooling unit sampled.

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Table 2. Compositions of Upper Cretaceous Ash Flow Tuffs in the Bell Mountain Area, Roskruge Mountains, Arizona

Sample	SiO_2	Al_2O_3	Fe_2O_3	MgO	CaO	Na_2O	K_2O	TiO_2	P_2O_5	MnO	LOI	Rb	Sr	Y	Zr	Nb	Ba	Ce
4b	77.4	13.3	0.75	0.21	0.17	3.50	4.51	0.05	0.03	0.00	1.62	340	37	50	101	45	46	29
4b (low)	77.1	13.4	0.69	0.17	0.04	4.14	4.26	0.06	0.03	0.13	0.87	277	40	56	98	43	41	27
4a (high)	77.4	13.6	0.74	0.11	0.01	5.12	2.90	0.07	0.03	0.03	1.00	117	56	24	78	22		
3b	73.1	15.6	2.43	0.00	0.10	3.23	4.92	0.42	0.03	0.04	0.43	153	127	33	375	17	1010	123
3a (base p)	73.8	12.4	1.91	0.26	3.23	2.22	5.73	0.32	0.05	0.10	3.23	191	136	21	280	16		
3a (base)	69.0	16.1	2.36	0.42	2.55	3.44	5.63	0.43	0.08	0.01	3.29	189	135	24	391	13		
3a (base)	67.7	14.3	2.25	0.39	4.43	5.20	5.07	0.39	0.03	0.12	4.63	184	151	40	354	17	1020	120
2 (upper)	74.7	14.1	0.94	0.39	1.86	3.14	4.64	0.11	0.03	0.13	2.94	187	153	17	80	16		
2 (middle)	77.2	12.9	0.95	0.23	0.24	5.20	3.17	0.12	0.03	0.00	1.25	133	176	19	92	19	387	42
1b (top)	74.7	13.6	1.83	0.51	0.90	4.75	3.40	0.23	0.03	0.03	1.16	95	258	17	96	14	1290	59
1b (base)	74.4	14.3	2.01	0.30	1.74	4.29	2.60	0.26	0.03	0.03	2.14	69	300	13	92	13	850	48
1a	74.6	14.7	2.09	0.61	1.80	3.06	2.75	0.26	0.03	0.04	1.91	78	329	14	90	11	812	44
Viopuli	68.9	14.5	2.15	0.31	2.32	5.59	5.69	0.39	0.03	0.06	2.77	246	102	41	378	16	1280	129

Major oxide analyses are reported as weight percent and are normalized to 100% volatile free; total iron reported as Fe_2O_3; analyzed using XRF methods by J. Taggart. Minor elements reported in parts per million; analyzed using nondispersive KEVEX method by D. Yager and D. Sawyer. LOI, loss on ignition; Viopuli, Viopuli Tuff; p, pumice sample.

Paleomagnetism

Oriented samples from the four tuffs exposed east of Bell Mountain (Figure 1) were subjected to stepwise demagnetization using either peak alternating fields (AF) between 5 and 100 mT or heatings up to 680°C. *Vugteveen et al.* [1981] corrected their sites on Bell Mountain (RV4–12) for strike and dip angles of N75°E, 19°SE, and we used an attitude of N60°E, 20°SE for our sites east of Bell Mountain (RT1–4, Table 3). Structural corrections of the homoclinal section were made by simple rotation of the bedding planes (eutaxitic structures) to horizontal about the strike axis. The demagnetization data were fitted with either least squares lines or planes [*Kirschvink*, 1980], and standard statistical methods [*Fisher*, 1953] were used in analyzing characteristic directions determined by line solutions; Bingham statistics [*Onstott*, 1980] were used in analyzing mean directions determined by intersecting demagnetization planes.

Directions of natural remanent magnetization (NRM) measured prior to demagnetization treatments were dispersed for each site, suggesting the presence of lightning-induced components. AF demagnetization (>20 mT) was successful in removing these secondary magnetizations in samples from tuff 3 (Figure 2a), and their characteristic magnetizations could be fitted by least squares lines. AF demagnetization to 100 mT, however, was generally unsuccessful in isolating characteristic magnetizations in samples from tuffs 1 and 4, but mean directions are clearly defined by intersecting demagnetization planes (Figures 2b and 2c).

Tuff 2 was initially collected (10 samples) at a site east of the San Pedro Road, and AF demagnetization was unable to resolve a characteristic direction of magnetization; up to ~25% of the NRM intensity remained after the 100-mT step. The tuff is lithic-rich and only partly welded, so the dispersion of NRM directions might be due to randomly oriented magnetizations of lithic fragments that were not sufficiently reheated and remagnetized by the tuff. Tuff 2 was resampled (19 samples) at a second site west of the San Pedro Road, and the available specimens (28 total) were subjected to progressive thermal demagnetization. The thermal demagnetization curves range from simple to complex, and the blocking temperatures suggest that both magnetite and hematite are present. Linear demagnetization curves for these samples have random directions and were discarded, but demagnetization planes for 15 samples apparently intersect near a common characteristic direction. Because the blocking temperature spectra of the lithic inclusions are unknown, demagnetization planes were accepted for data points that moved progressively toward or away from the unit mean direction (Figure 2d).

Table 3. Paleomagnetic Data for Upper Cretaceous Ash Flow Tuffs at Bell Mountain, Arizona

Sites	I	D	N	R	k	α_{95}	λ	ϕ
				Tuff 4b				
RV12*	61.4	348.4	5	4.98	192	5.5	76.1	210.5
RV11†	68.8	6.4	4	3.88	25	18.6	69.3	259.7
				Tuff 4a				
RT4	−42.2	161.1	7		[156, 2]	[3.6, 5.7]	71.7	138.5
RV10	−47.2	156.8	4	3.99	503	4.1	69.7	154.1
RV9	−47.5	151.7	4	3.99	231	6.1	65.4	157.9
RV8	−49.5	155.3	5	4.97	142	6.4	68.9	160.4
RV7	−51.6	152.1	5	5.00	1163	2.2	66.5	166.5
RV6	−53.1	155.5	4	3.98	175	7.0	69.4	169.6
Avg*	−48.6	155.6	(6)	5.9846	324	3.7	68.9	158.1
				Tuff 3				
RT3b	46.5	359.2	9	8.9875	639	2.0	85.7	77.6
RT3a	44.4	6.2	9	8.9816	435	2.5	82.0	24.9
Avg*	45.5	2.8	2	1.9978	461	11.7	84.4	42.5
				Tuff 2				
RT2	−65.0	270.1	15		[39, 1]	[5.5, 6.5]	22.8	296.3
				Tuff 1 (a or b)				
RT1*	−65.1	127.2	12		[132, 1]	[3.2, 4.0]	47.5	195.1
RV5†	−60.4	161.2	5	4.90	41	12.0	72.4	195.6
				Viopuli Tuff				
RV4*	50.1	354.5	7	6.98	271	3.7	85.2	145.8

RV sites from *Vugteveen et al.* [1981], RT sites from this study; I and D are inclination and declination of paleomagnetic directions (degrees) corrected for structural tilt; N is number of samples (sites) analyzed; R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} is the radius of 95% confidence (degrees). For planes solutions, R is not calculated, $[-k_1, -k_2]$ are the two Bingham concentration parameters, and $[\alpha_1, \alpha_2]$ are the two Bingham 95% confidence radii [*Onstott*, 1980]. Here λ and ϕ are north latitude and east longitude of virtual geomagnetic pole (degrees). Average site locality, 32.06°N, 248.51°E.

*Included in overall mean.

†Larger dispersion probably caused by lightning strikes.

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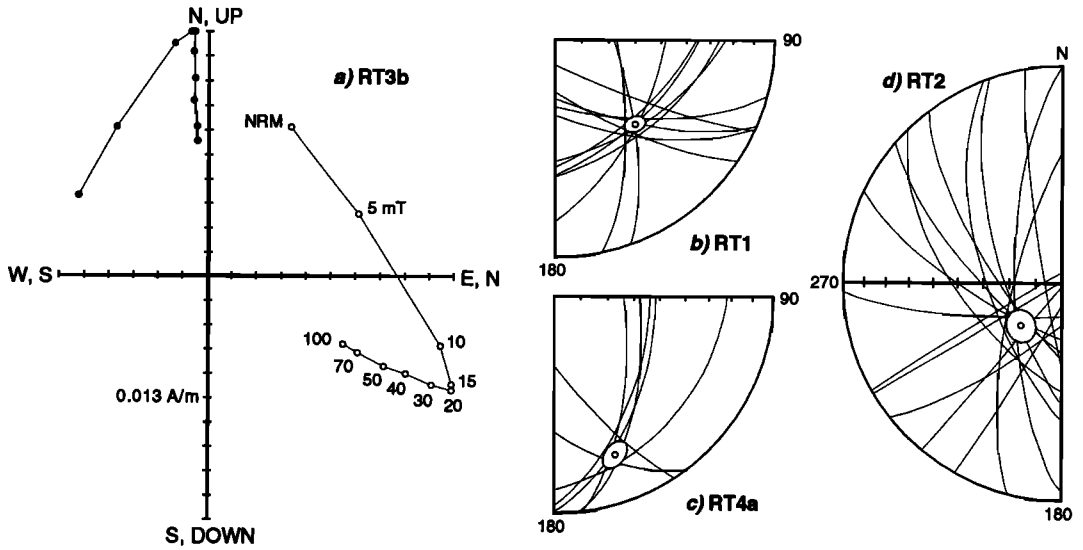


Figure 2. Orthogonal projection of alternating field demagnetization data, uncorrected for structural tilt, for a sample of (a) ash flow 3b showing removal of a lightning-induced component of magnetization by 20 mT; solid (open) symbols indicate projection onto the horizontal (vertical) plane. Equal-area plots showing converging demagnetization planes and unit mean directions with 95% confidence limits, uncorrected for tilt, for (b) ash flow 1, (c) ash flow 4a, and (d) ash flow 2. Open symbols indicate projection from the upper hemisphere.

Discussion

Ash Flow Tuff Correlations

Site mean directions for the tuffs at Bell Mountain are given in Table 3 and shown in Figure 3, including those determined by *Vugteveen et al.* [1981]. The lowermost red Viopuli Tuff (RV4) has a distinct appearance in the field and was not resampled for this study; it has a normal polarity direction of magnetization. Tuff 1 (RT1) has a reversed polarity direction, and *Vugteveen et al.* [1981] apparently sampled this tuff at site RV5 on Bell Mountain. Their mean direction is somewhat different from ours and shows higher dispersion, probably due to incomplete removal of lightning-induced magnetizations in these samples. Tuff 2 (RT2) also has a reversed polarity magnetization with an unusual, perhaps transitional, direction; the two sites discarded by *Vugteveen et al.* [1981] due to poor demagnetization behavior were probably from this weakly welded lithic-rich tuff. Tuffs 3a and 3b (RT3a–b) show similar normal polarity directions of magnetization, and *Vugteveen et al.* [1981] appear not to have sampled this unit. Tuff 4a (RT4) has a reversed polarity magnetization, and *Vugteveen et al.* [1981] apparently sampled this tuff at five sites (RV6–10). The uppermost site (RV12), collected by *Vugteveen et al.* [1981] near the summit of Bell Mountain, has a normal polarity direction suggesting the presence of an additional ash flow sheet, tuff 4b. Site RV11 shows a similar mean direction and is most likely from the same unit, but its higher dispersion is also probably caused by lightning strikes.

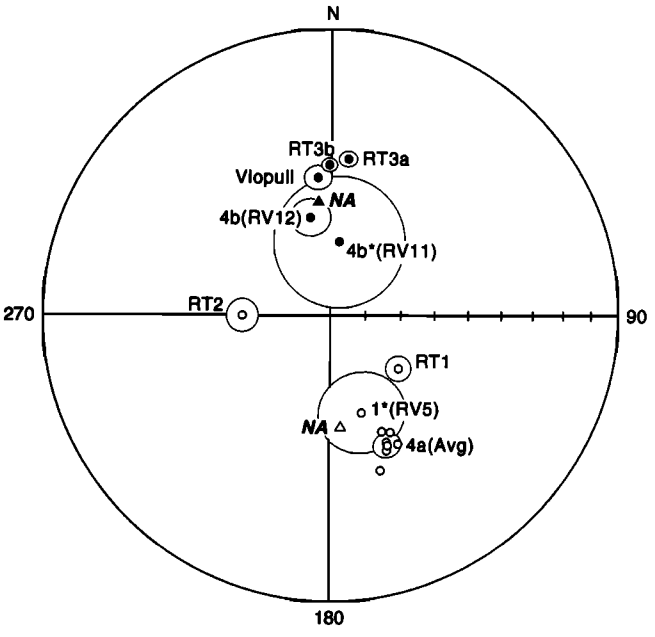


Figure 3. Equal-area plot of mean paleomagnetic directions with 95% confidence limits, corrected for structural tilt, for ash flow tuffs (1–4) on Bell Mountain (see Table 2). Viopuli, mean direction for the underlying Viopuli Tuff; *NA*, expected Late Cretaceous dipolar field direction (triangles) for Bell Mountain. Asterisk denotes site means of higher dispersion probably affected by lightning strikes. Solid (open) symbols indicate projection from the lower (upper) hemisphere.

The interpretation of an additional ash flow unit capping Bell Mountain (tuff 4b) requires that our field interpretation of an incomplete cooling break within tuff 4 is mistaken. A troubling aspect of such an attempt to relate the data of *Vugteveen et al.* [1981] to the sequence we have sampled is that inference of an additional tuff (4b) would also imply that they neglected sampling tuff 3 (the most prominent cliff-forming tuff), placed five separate sampling sites within a single 50-m-thick cooling unit (our tuff 4a), and misinterpreted these sites as sampling five separate cooling units. Until further sampling can be undertaken on Bell Mountain, our correlations with the data of *Vugteveen et al.* [1981] will remain uncertain.

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Although tuff 1 appears compositionally similar to the normal polarity Confidence Peak Tuff (~73 Ma) in the Silver Bell Mountains [*Sawyer*, 1989], it has a reversed polarity direction and cannot be the same tuff [*Hagstrum and Sawyer*, 1989]. Because of its very low phenocryst content and abundant lithic fragments, tuff 2 is unlike either the Confidence Peak Tuff or Cat Mountain Tuff (~72 Ma) in the Tucson Mountains [*Lipman*, 1993]. Tuff 3 has a high Zr content (~370 ppm) like the Cat Mountain Tuff (>100–400 ppm), but it completely lacks quartz phenocrysts and has normal polarity. The reversed polarity and similar inclination (−49° ± 4° versus −55° ± 3°) for tuff 4 also make it a candidate for Cat Mountain Tuff [*Hagstrum and Lipman*, 1991], but its chemistry is different (Zr < 100 ppm; Fe_2O_3 total ~0.75% versus ~1.85% for the Cat Mountain Tuff).

Based on broad lithologic similarity, Roskruge tuff units 2 and 4 are permissible correlatives with some of the six currently recognized tuff units of the Red Boy Rhyolite erupted from the Sierrita caldera 30 km to the southeast of Bell Mountain [*Lipman and Fridrichs*, 1990]. Available chemical and petrographic data are inadequate for detailed comparisons of the Red Boy Rhyolite, which is preserved only within the Sierrita caldera. Alternatively, the multiple flow units, interleaved surge bedding, and complex welding zonations of Bell Mountain tuffs 2 and 4 suggest closer proximity to a eruptive source, and the Cocoraque Butte Granodiorite in the Roskruge Mountains (also on tribal land) could be a subvolcanic intrusion related to at least some of the multiple cooling unit tuffs in the Bell Mountain area.

Reliability of Upper Cretaceous Paleopoles

Vugteveen et al.'s [1981] paleopole has been among the few available for Late Cretaceous North America and has been assigned a quality factor of 5 (out of 7) by *Van der Voo* [1993]. Two quality criteria were failed in this analysis because field stability tests are lacking and the tectonic coherence of southern Arizona since the Cretaceous has not been established. The Roskruge Volcanics are unaltered and show reversals implying primary magnetizations and were analyzed by *Vugteveen et al.* [1981] using modern techniques with good precision. Our results, however, indicate that they sampled only four time-independent tuffs instead of nine, as reported. Therefore their paleopole likely does not include sufficient data to average out paleosecular variation (PSV) of the geomagnetic field. This was also suggested by *Vugteveen et al.* [1981], because their normal and reversed polarity directions are not antipodal. An overall mean direction for Bell Mountain (Table 3), omitting unit 2 and averaging units 3a and 3b, is $I = 55.5°$, $D = 344.6°$, $N = 5$, $k = 32$, and $\alpha_{95} = 13.8°$. This direction is similar to *Vugteveen et al.*'s [1981] mean direction and a North American reference direction ($I = 57.4°$, $D = 353.8°$, $\Delta I = 5.0°$, and $\Delta D = 6.7°$; calculated from a paleopole for the Adel Mountains Volcanics (~71–81 Ma) given by *Gunderson and Sheriff* [1991]) but also has a small N and probably does not average out PSV.

Other Late Cretaceous paleopoles for North America [see *Van der Voo*, 1993] are from the Boulder batholith (~70–80 Ma) [*Hanna*, 1973], Elkhorn Mountains Volcanics (~75–80 Ma) [*Hanna*, 1967, 1973; *Diehl*, 1991], Niobrara Formation (~69–88 Ma) [*Shive and Frerichs*, 1974; *Acton and Gordon*, 1990], and Maudlow Formation (~75–83 Ma) [*Swenson and McWilliams*, 1989]. *Van der Voo* [1993] discards the Boulder batholith result because there is no structural control for these rocks, and tilting is suspected. *Diehl* [1991] determined a new pole for the Elkhorn Mountains Volcanics, different from the old pole of *Hanna* [1967, 1973], that is similar to Paleocene reference poles [*Diehl et al.*, 1983]. The Elkhorn Mountains Volcanics, however, are floored by a major east directed thrust which may have affected these data. *Gunderson and Sheriff* [1991] consider the Maudlow Formation pole unreliable because it is from rocks within the Helena salient of the fold and thrust belt, which lie near (<20 km) Cretaceous rocks that have been rotated 24°–41° about near-vertical axes [*Eldredge and Van der Voo*, 1988]. There are also problems associated with poles for the Niobrara Formation. Recently, *Acton and Gordon* [1990] have determined a second pole for the Niobrara Formation (75°N, 185°E, $A_{95} = 3°$) that is significantly different from *Shive and Frerichs*' [1974] pole (66°N, 192°E, $A_{95} = 6°$). In addition, ages of the Niobrara Formation members sampled by *Shive and Frerichs* [1974] are uncertain, and their pole is assigned older ages (85–89 Ma) by *Globerman and Irving* [1988]. *Gunderson and Sheriff* [1991] sampled the Adel Mountains Volcanics (~71 to 81 Ma) to the east of the fold and thrust belt in Montana and apparently obtained a reliable pole similar to Paleocene reference poles [*Diehl et al.*, 1983]. These data, however, are only of normal polarity, and PSV therefore might not be adequately averaged.

Conclusions

Our mapping and petrologic studies of Upper Cretaceous rocks in the Bell Mountain area have identified five compositionally distinct ash flow tuffs comprising six or seven separate cooling units. Comparisons of our characteristic directions of magnetization for these units with *Vugteveen et al.*'s [1981] results from Bell Mountain indicate that they sampled only four independent tuff units. Their resultant paleopole therefore likely does not average PSV of the geomagnetic field and should not be considered an accurate Late Cretaceous pole for North America. Few other poles are available for this time and that for the ~76-Ma Adel Mountains Volcanics [*Gunderson and Sheriff*, 1991] appears most reliable.

Because no correlatable tuffs were found at Bell Mountain and comparisons of declination data cannot be made, relative vertical axis rotations for the locality are indeterminate. The similarity of observed and expected directions, however, implies that any such rotations since the Late Cretaceous are probably not large. The lack of ash flow sheets at Bell Mountain that have been previously identified elsewhere implies the presence of additional, potentially mineralized, volcanic centers in the region [see *Lipman and Sawyer*, 1985]. The Roskruge Mountains themselves are one probable locality, and others may lie buried nearby beneath Cenozoic basin fill.

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